

GATE Question Paper 2010, EE Question Number 54

Abstract

54. The initial current through the inductor is zero, while the initial capacitor voltage is 100 V. The switch is closed at $t = 0$. The current i through the circuit is:
- (A) $5 \cos(5 \times 10^3 t)$ A (B) $5 \sin(10^4 t)$ A
(C) $10 \cos(5 \times 10^3 t)$ A (D) $10 \sin(10^4 t)$ A

(GATE 2010, Question No. 54 – Natural Response of LC Circuit: Simulated on Raspberry Pi Pico)

1. Components Used

Component	Qty
Raspberry Pi Pico (RP2040)	1
Breadboard	1
LED	1
220Ω Resistor	1
Jumper Wires (M-M)	6
USB Cable (Micro-B)	1
Computer with Thonny IDE	1

Table 1: List of components used

2. Setup and Connections

1. Connect GP15 (PWM pin) to LED through 220Ω resistor.
2. Connect cathode of LED to GND.
3. Connect Raspberry Pi Pico to PC using USB cable.
4. Flash MicroPython firmware and open Thonny IDE.

3. Simulation Strategy

- Generate a cosine waveform: $i(t) = 10 \cos(5000t)$.
- Use MicroPython to simulate waveform using PWM.
- LED brightness varies according to the waveform.

5. Working Principle

- The waveform simulates current in an LC circuit.
- The LED acts as a visual indicator of waveform magnitude.
- Cosine waveform is generated using a lookup table.
- 100 samples/cycle approximates the waveform smoothly.

6. Observations and Output

- LED brightness oscillates sinusoidally.
- Represents undamped oscillations in LC circuit.
- Waveform can be viewed on oscilloscope from GP15.

7. Result The waveform $i(t) = 10 \cos(5000t)$ is successfully emulated using PWM on Raspberry Pi Pico. It matches the expected LC circuit response.

8. Source Code Link

https://github.com/madhuu34/Madhulathafwc/blob/main/Hardware/AVR_GCC/hardware_code.py